

STAT 7900: SPECIAL TOPIC: CAUSAL ANALYSIS



INSTRUCTOR

DHRUBAJYOTI GHOSH

KENNESAW STATE UNIVERSITY

STAT 7900

- Introduce yourself.
- Your program and year
- One thing you hope to learn from this course

- Class Structure: In-person and online
- Exams: 1 midterm and 1 final project.
- Assignments: 3 over the semester
- Quizzes: Daily quizzes on previous class content.

Causal Analysis

Examples of causal questions:

Medicine

Would a new treatment reduce mortality?

Education

Would reducing a class size improve achievement?

Policy

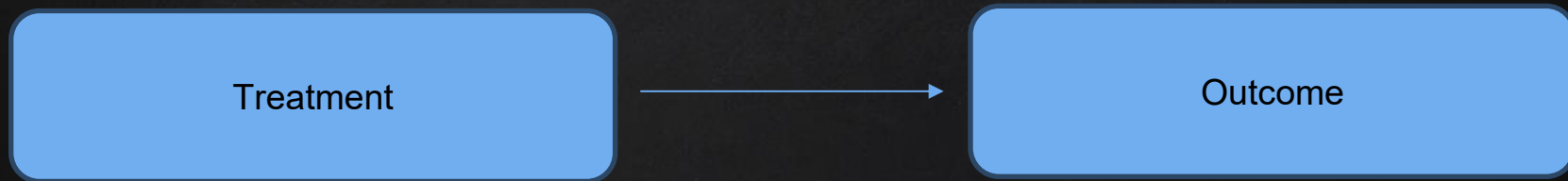
Would raising a minimum wage affect employment?

Public Health

Would reducing air pollution lower asthma admissions?

What is Causal Inference

Learning the effect of a treatment, exposure, policy, or intervention on an outcome.



Three Kinds of questions

Descriptive

What is happening?

Example: How does blood pressure differ between smokers and non-smokers?

Predictive

What will happen?

Example: Can age, BMI or smoking predict future blood pressure?

Causal

What would happen if we intervened?

Example: Would smoking cessation lower blood pressure?

Three Kinds of questions

We can have same dataset but different questions.

Imagine a cohort with diet, age, BMI, blood pressure, and future cardiovascular events.

Describe: What is the average sodium intake?

Predict: Who will develop hypertension?

Cause: What would happen to blood pressure if sodium intake were reduced?

The scientific question determines the analysis.

Simpson's Paradox

New disease: COVID-27.

Treatment T: A (0), B (1).

You need to decide which treatment to use for the country.

Condition C: mild (0), severe (1)

Outcome Y: alive (0), dead (1)

Simpson's Paradox: Mortality Rate Table

Treatment		Total
	A	16% (240/1500)
	B	19% (105/550)
$\mathbb{E}[Y T]$		

Simpson's Paradox: Mortality Rate Table

		Condition		
		Mild	Severe	Total
Treatment	A	15% (210/1400)	30% (30/100)	16% (240/1500)
	B	10% (5/50)	20% (100/500)	19% (105/550)

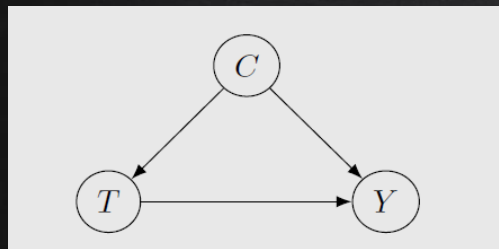
$\mathbb{E}[Y|T, C = 0]$

$$\frac{1400}{1500} (0.15) + \frac{100}{1500} (0.30) = 0.16$$

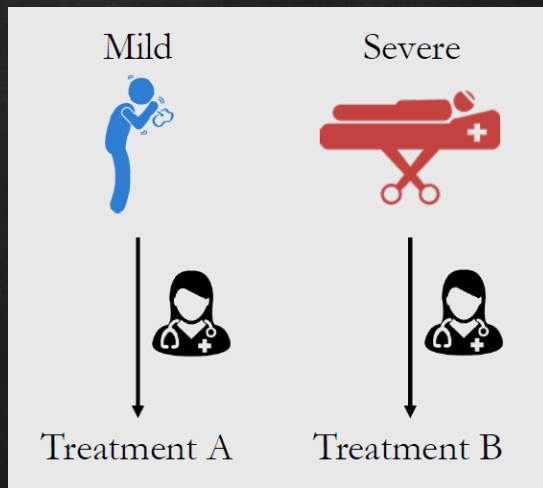
;

$$\frac{50}{550} (0.10) + \frac{500}{550} (0.20) = 0.19$$

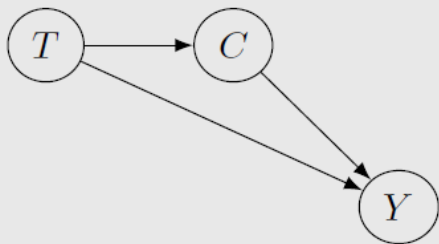
Simpson's Paradox: Which treatment is better?



Treatment B is
better



Simpson's Paradox: Which treatment is better?



Treatment A is
better

Treatment A



Mild

Treatment B



Severe

Simpson's Paradox: A Real dataset example

R includes the classic 1973 Berkeley graduate admissions table as `UCBAdmissions`.

- 4526 Applicants
- 6 Departments
- 2 gender categories
- 2 Admission Outcomes

Question: Were female applicants less likely to be admitted?

Simpson's Paradox: A Real dataset example

Overall Admissions Rate:

Male applicants: 44.5% (1198 admitted out of 2691 applicants)

Female applicants: 30.4% (557 admitted out of 1835 applicants)

What would you conclude?

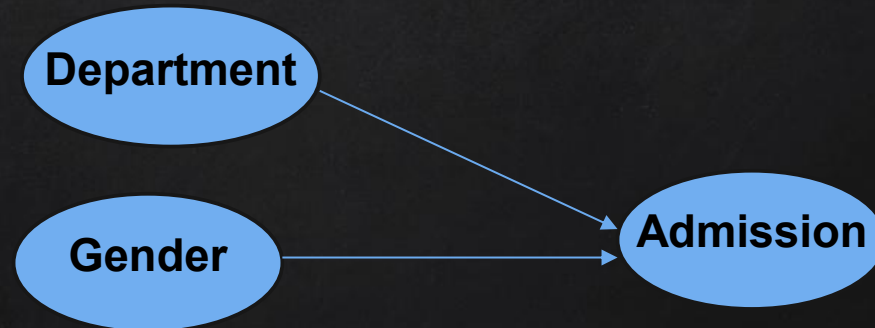
Does 44.5% vs 30.4% show that being female reduced the chance of admission?

Simpson's Paradox: A Real dataset example

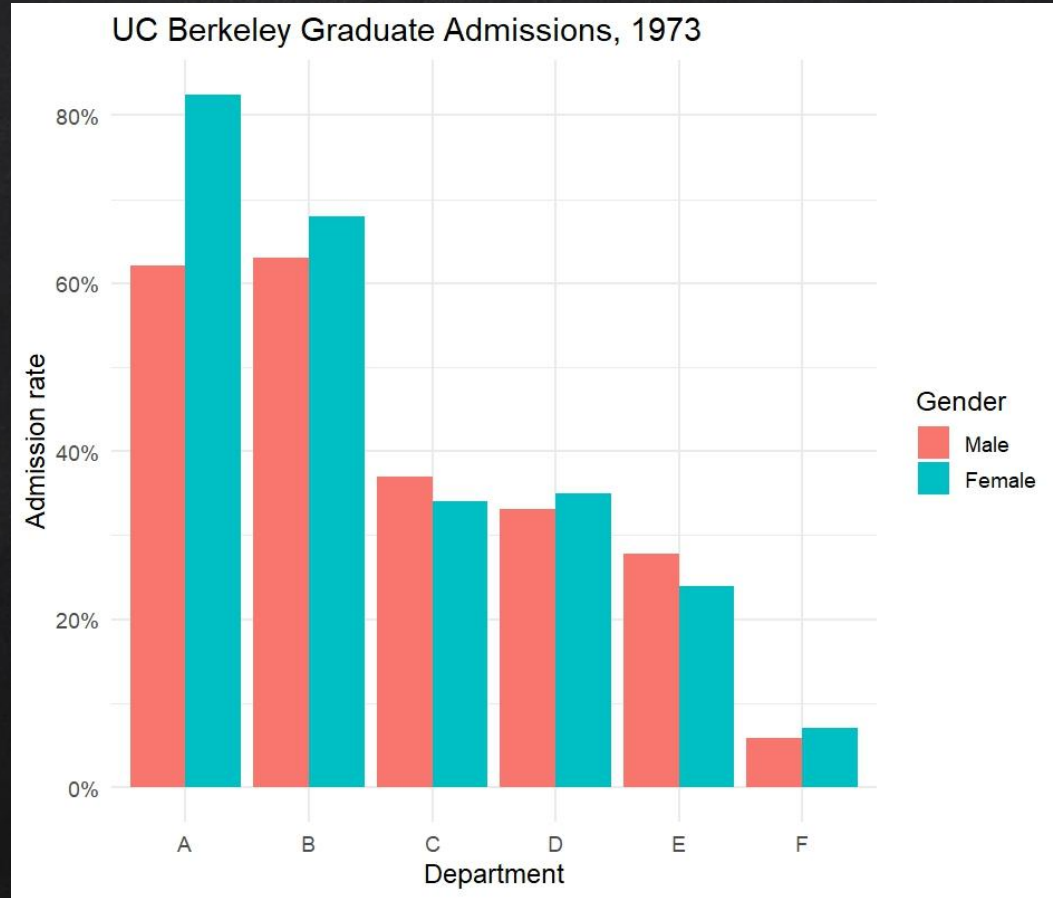
A hidden structure: Department

Applicants did not apply to the same department in the same proportions.

Departments also have different selectivity.



Simpson's Paradox: A Real dataset example



Simpson's Paradox: A Real dataset example

The aggregate comparison and the within group comparisons can point in different directions.

Overall: Male 44.5%, Female 30.4%

When we consider within departments, the gap shrinks or reverses in several departments.

- Departments had very different admission rates.
- Male and female applicants applied to those departments in different proportions.
- The overall rate is a weighted average of department-specific rates.
- Changing the weights can change the apparent association.

Correlation and Causation

Correlation does not imply causation.

Example 1: Sleeping with shoes on is strongly correlated with waking up with a headache.
Common Cause: Drinking the night before.

Shoe-sleepers differ from non-shoe sleepers in a key way.
This is called **Confounding**.

Total association (e.g. Correlation): Mixture of causal and confounding association.

“Correlation = Causation” is a cognitive bias.

Correlation and Causation

Different types of cognitive bias:

- Availability heuristic
- Motivated Reasoning
- Anchoring bias
- Confirmation bias
- Apophenia

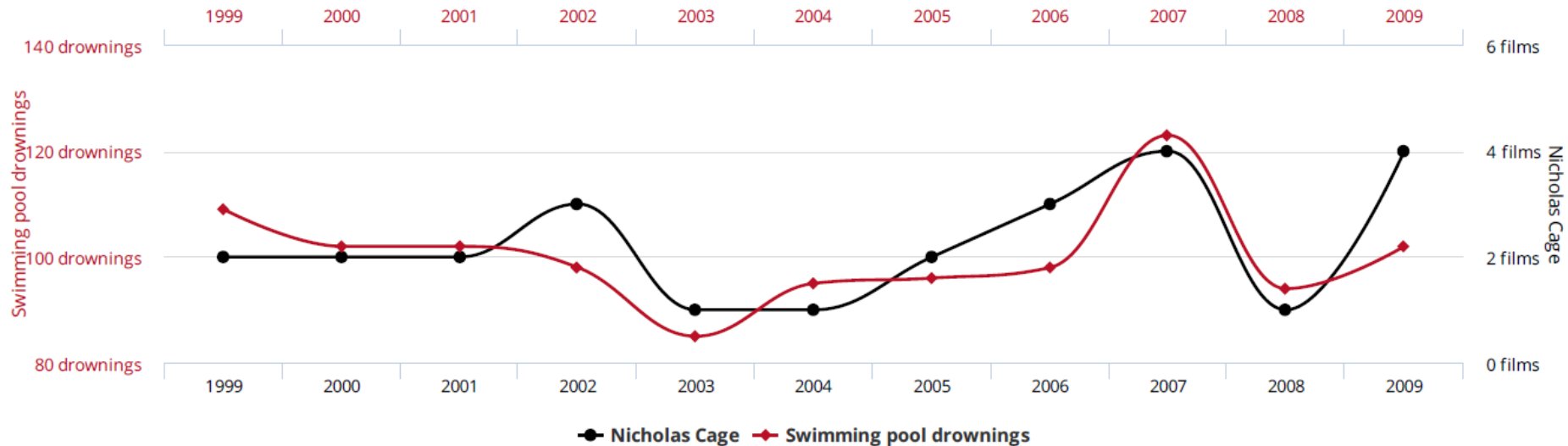
Correlation and Causation

Nicolas Cage drives people to drown themselves

Number of people who drowned by falling into a pool

correlates with

Films Nicolas Cage appeared in



Prediction and Causation

Prediction and Causation answer two different questions.

Prediction: $P(Y | X)$.

Causation: $E(Y(1) - Y(0))$.

Suppose an electronic health record contains age, diagnosis, labs, treatment, and mortality.

Prediction: Which patients are most likely to die?

Causation: Would giving treatment A reduce mortality?

Treatment itself may be a strong predictor because the sickest patients receive it.

Prediction and Causation

High predictive accuracy does not establish a causal effect.

A model can perfectly learn patterns that are consequences of confounding, selection mechanisms, treatment decisions, post-treatment variables, etc.

Prediction asks whether the pattern repeats.

Causal inference asks what would change under intervention.

Formulating a causal question

Can I meaningfully imagine changing the exposure?

Association: Are smokers less healthy?

Causal: What would happen to health if smokers stopped smoking?

Association: Are highly educated people wealthier?

Causal: What would earnings be if a person received one additional year of education?

Association: Do people who exercise more have lower blood pressure?

Causal: What would happen to blood pressure if sedentary adults increased their exercise by 150 minutes per week?

Association: Are people who drink coffee less likely to develop Parkinson's disease?

Causal: What would happen to Parkinson's disease risk if people increased their coffee consumption?

Formulating a causal question

Five ingredients of a causal question

1

Population

Who?

2

Treatment

What
intervention?

3

Comparator

Compared with
what?

4

Outcome

What
endpoint?

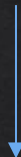
5

Time

Over what
horizon?

Formulating a causal question

Does tutoring help students?



Among students eligible for tutoring, what would be the difference in pass rates if all were assigned tutoring versus if none were assigned tutoring?

What implies causation?



Take pill



causal effect

Don't take pill



What implies causation?

Inferring the effect of treatment/policy on some outcome



Take pill



Don't take pill



no causal effect

Potential Outcomes

Suppose T is the treatment (=1 means treated, =0 means not treated).

Y : Observed outcomes, i is used in subscript to denote a specific unit/ individual.

Take the pill: $do(T = 1)$, Outcome: $Y_{i|do(T=1)} \equiv Y_i(1)$

Do not take the pill: $do(T = 0)$, Outcome: $Y_{i|do(T=0)} \equiv Y_i(0)$

$Y_i(1)$ is **potential outcome** under treatment.

$Y_i(0)$ is **potential outcome** under no treatment.

Causal Effect: $Y_i(1) - Y_i(0)$

Potential Outcomes

World 1

Treatment Received

$$A = 1$$

Outcome: $Y(1)$

World 2

No treatment

$$A = 0$$

Outcome: $Y(0)$

Potential Outcomes: Fundamental Problem

You cannot observe both $Y_i(1)$ and $Y_i(0)$ at the same time.

You cannot be given treatment and not given treatment at the same time.

Example: If you are given treatment, you can only observe $Y_i(1)$.

$Y_i(0)$ cannot be observed, and this is called counterfactual.

If $A=1$, observed $Y(1)$. Then $Y(0)$ is counterfactual.

If $A=0$, observed $Y(0)$. Then $Y(1)$ is counterfactual.

Causal inference is the science of learning about those missing counterfactuals.

Average Treatment Effect

T: Observed Treatment

Y: Observed Outcome

i : Used in subscript to denote a specific individual

$Y_i(1)$: Potential Outcome under treatment

$Y_i(0)$: Potential Outcome under no treatment

$Y(t)$: Population-level potential outcome

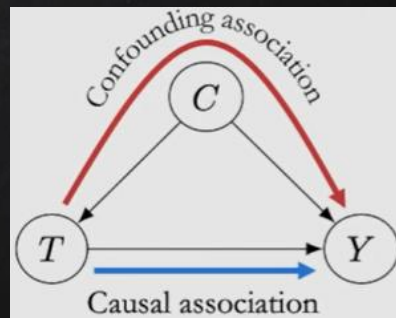
Individual Treatment Effect (ITE): $Y_i(1) - Y_i(0)$

Average Treatment Effect (ATE):

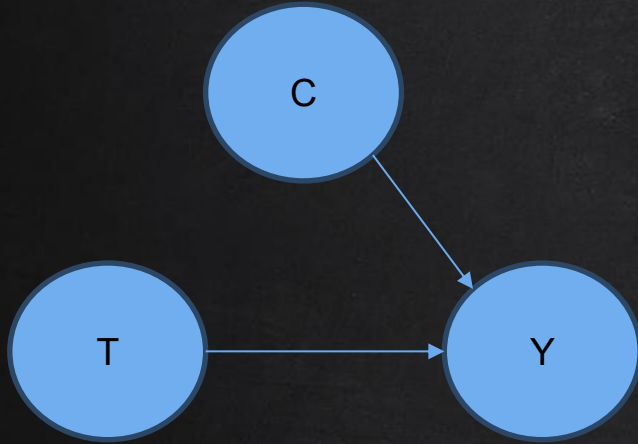
$$E[Y_i(1) - Y_i(0)] = E(Y(1)) - E(Y(0))$$

$$\neq E(Y|T=1) - E(Y|T=0)$$

This is because Correlation is not equal to Causation.



Average Treatment Effect: Randomized Control Trials



Under RCT, association can mean causation, if all confounding factors are appropriately taken care of.

Under confounding, we saw that ATE is not same as conditional expectation.

$$E[Y(1) - Y(0)] \neq E[Y|T = 1] - E[Y|T = 0]$$

However, this is true for RCTs.

RCT: Experimenter randomizes subjects into treatment groups or control group.

T cannot have any causal parents, which means groups are comparable.

Example: A drug is tested on a group of patients. The drug is administered approximately equally on males and females, so that there is no confounding due to age.

Average Treatment Effect: Observational Studies

Randomization of treatment is not always possible.

Ethical

Unethical to
randomize people to
smoke for
measuring effect on lung
cancer

Infeasible

Cannot randomize
countries into
communist/capitalist
systems to measure effect
on GDP

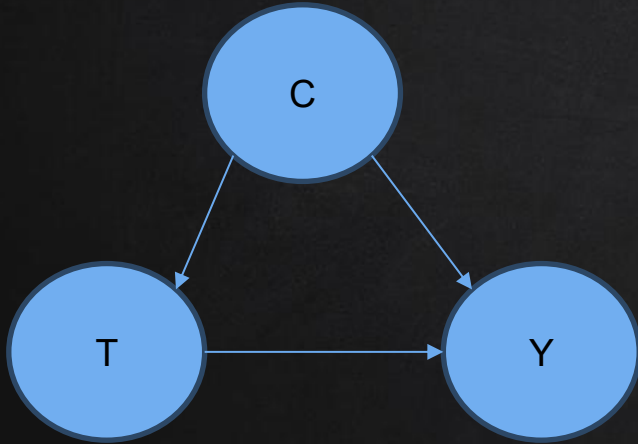
Impossible

Cannot change a living
person's DNA at birth for
measuring
effect on breast cancer

Average Treatment Effect: Observational Studies

Most real-world causal questions require observational causal inference.

Observational causal inference: When treatment is not randomized, treatment assignment itself carries information.



The central question becomes:

Can we recover the causal comparison using assumptions + observed data?

A simulated Experiment

In simulation we can observe both potential outcomes.

We construct a world such that $Y(1) - Y(0) = -3$.

X is a common cause of treatment and the outcome.

Crude $E(Y|A = 1) - E(Y|A = 0)$ does not give the true causal effect.

Adjustment for the right variable recovers the causal effect.

